The Short Answer
The PDF viewer is guessing there is a space because the physical gap between the letters i and c is slightly too wide.

Table full width

The Technical Explanation

PDFs are Coordinate Maps, Not Text Streams: Unlike a Word document, a PDF does not store sentences or words. It stores instructions like: "Draw glyph 'Sacrifi' at x=100, y=200" and "Draw glyph 'ce' at x=130, y=200."

Kerning & Justification: To make the text look smooth or fit a specific width (justification), the generator (iText 5.5.4) likely adjusted the spacing between specific characters.

Heuristic Failure: When you copy text, the PDF viewer measures the distance between the end of "Sacrifi" and the start of "ce." If that gap exceeds a specific threshold (e.g., 20% of the font size), the viewer's extraction algorithm assumes a space character exists there, inserting it into your clipboard.

Why iText 5.5.4 Matters
The metadata shows the file was generated programmatically by a library (iText), not typed manually in a word processor. Programmatic generation often paints text in fragments to handle complex layout logic, increasing the likelihood of these "coordinate gaps" occurring in the middle of words.
Numerous concepts are related to this idea, at least one of which predates Goodhart's statement.[4] Notably, Campbell's law likely has precedence, as Jeff Rodamar has argued, since various formulations date to 1969.[5] Other academics had similar insights at the time. Jerome Ravetz's 1971 book Scientific Knowledge and Its Social Problems[6] also predates Goodhart, though it does not formulate the same law. Ravetz discusses how systems in general can be gamed, focusing on cases where the goals of a task are complex, sophisticated, or subtle. In such cases, the persons possessing the skills to execute the tasks properly seek their own goals to the detriment of the assigned tasks. When the goals are instantiated as metrics, this could be seen as equivalent to Goodhart and Campbell's claim.

Shortly after Goodhart's publication, others suggested closely related ideas, including the Lucas critique (1976). As applied in economics, the law is also implicit in the idea of rational expectations, a theory in economics that states that those who are aware of a system of rewards and punishments will optimize their actions within that system to achieve their desired results. For example, if an employee is rewarded by the number of cars sold each month, they will try to sell more cars, even at a loss.

While it originated in the context of market responses, the law has profound implications for the selection of high-level targets in organizations.[3] Jon Danielsson states the law as

Any statistical relationship will break down when used for policy purposes.

And suggested a corollary for use in financial risk modelling:

Gart's Law' – That every measure which becomes a target becomes a bad measure – is inexorably, if ruefully, becoming recognized as one of the overriding laws of our times. Ruefully, for this law of unintended consequences seems so inescapable. But it does so, I suggest, because it is the inevitable corollary of that invention of modernity: accountability.[10][full citation needed]

table

In a 1997 paper on the misuse of accountability models in education, anthropologist Marilyn Strathern cited Hoskins expressing Goodhart's Law as "When a measure becomes a target, it ceases to be a good measure", and linked the sentiment to the history of accountability stretching back into Britain in the 1800s:

When a measure becomes a target, it ceases to be a good measure. The more a 2.1 examination performance becomes an expectation, the poorer it becomes as a discriminator of individual performances. Hoskin describes this as 'Goodhart's law', after the latter's observation on instruments for monetary control which led to other devices for monetary flexibility having to be invented. However, targets that seem measurable become enticing tools for improvement. The linking of improvement to commensurable increase produced practices of wide application. It was that conflation of 'is' and 'ought', alongside the techniques of quantifiable written assessments, which led in Hoskin's view to the modernist invention of accountability. This was articulated in Britain for the first time around 1800 as 'the awful idea of accountability' (Ref. 3, p. 268).[1]

Examples

The San Francisco Declaration on Research Assessment denounces several problems in science and as Goodhart's law explains, one of them is that measurement has become a target. The correlation between h-index and scientific awards is decreasing since widespread usage of h-index.[11]

Mario Biagioli related the concept to consequences of using citation impact measures to estimate the importance of scientific publications:[8][9]

All metrics of scientific evaluation are bound to be abused. Goodhart's law [...] states that when a feature of the economy is picked as an indicator of the economy, then it inexorably ceases to function as that indicator because people start to game it.

Generalization

Later writers generalized Goodhart's point about monetary policy into a more general adage about measures and targets in accounting and evaluation systems. In a book chapter published in 1996, Keith Hoskin wrote:

'Goodh